

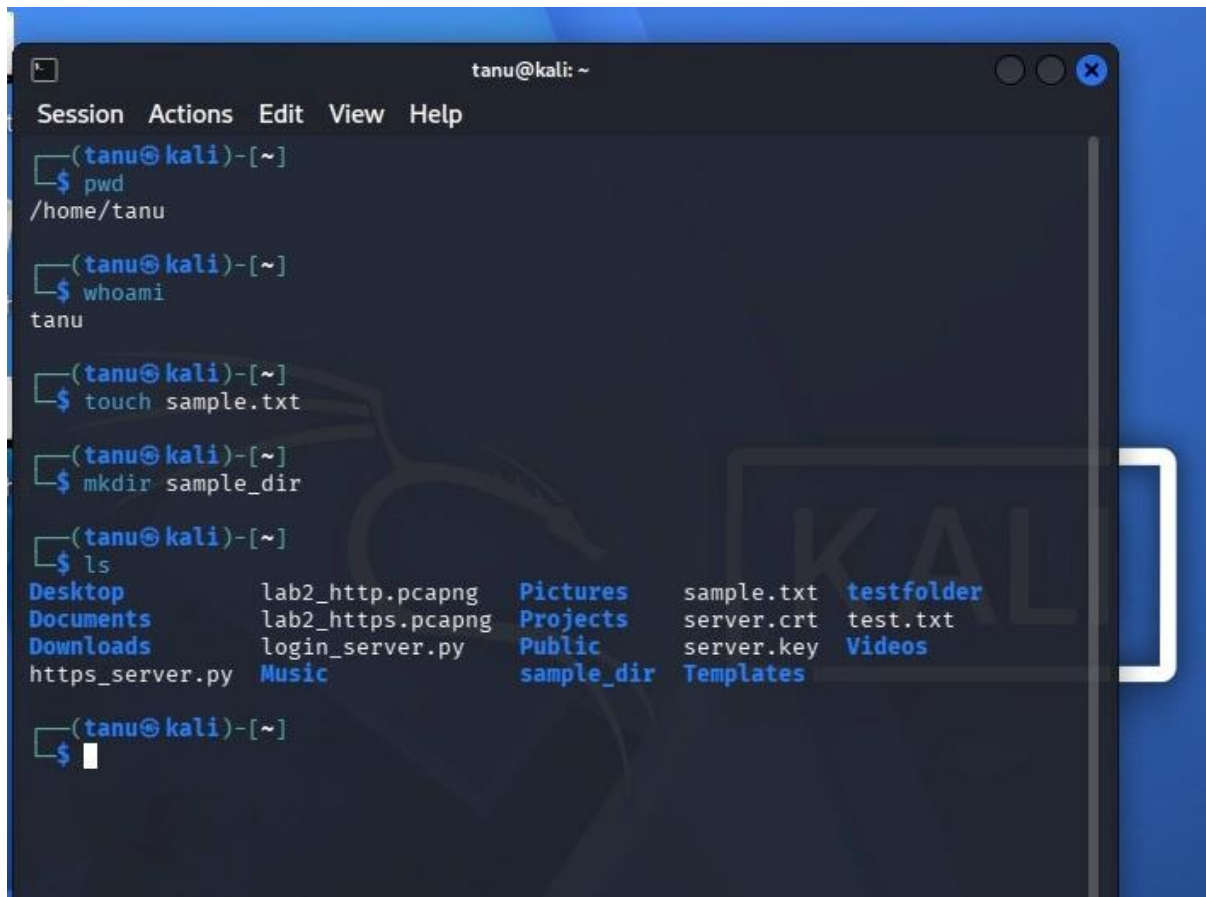
LOS Theory assignment – 2

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Compare Permissions on Regular Files and Directories

Step 1:



```
tanu@kali: ~  
Session Actions Edit View Help  
(tanu@kali)-[~]  
$ pwd  
/home/tanu  
(tanu@kali)-[~]  
$ whoami  
tanu  
(tanu@kali)-[~]  
$ touch sample.txt  
(tanu@kali)-[~]  
$ mkdir sample_dir  
(tanu@kali)-[~]  
$ ls  
Desktop          lab2_http.pcapng  Pictures          sample.txt        testfolder  
Documents         lab2_https.pcapng Projects          server.crt        test.txt  
Downloads         login_server.py   Public            server.key        Videos  
https_server.py  Music             sample_dir       Templates  
(tanu@kali)-[~]  
$
```

Created the sample.txt file

Created the sample_dir

Created a file inside the sample.dir file1.txt

1.

Observation table :

Permission	Regular File – Test	Observation	Directory – Test	Observation
400 (r--)	Read and modify	Read allowed; modification denied	List and enter	Listing and entering denied

Screenshot:

```
tanu@kali: ~  
Session Actions Edit View Help  
└─$ ls -l sample_dir  
total 0  
-rw-rw-r-- 1 tanu tanu 0 Aug 21 00:10 file1.txt  
  
└─(tanu@kali)-[~]  
└─$ chmod 400 sample.txt  
  
└─(tanu@kali)-[~]  
└─$ ls -l sample.txt  
-r----- 1 tanu tanu 0 Aug 21 00:06 sample.txt  
  
└─(tanu@kali)-[~]  
└─$ cat sample.txt  
  
└─(tanu@kali)-[~]  
└─$ echo "hello" >> sample.txt  
zsh: permission denied: sample.txt  
  
└─(tanu@kali)-[~]  
└─$ chmod 400 sample+dir  
chmod: cannot access 'sample+dir': No such file or directory  
  
└─(tanu@kali)-[~]  
└─$ chmod 400 sample_dir  
  
└─(tanu@kali)-[~]  
└─$ ls -ld sample_dir  
dr----- 2 tanu tanu 4096 Aug 21 00:10 sample_dir  
  
└─(tanu@kali)-[~]  
└─$ ls sample_dir  
ls: cannot access 'sample_dir/file1.txt': Permission denied  
file1.txt  
  
└─(tanu@kali)-[~]  
└─$ cd sample_dir  
cd: permission denied: sample_dir  
  
└─(tanu@kali)-[~]  
└─$
```

For a directory:

- r = permission to **read the directory's list of names**
- x = permission to **enter/access the directory**
- w = permission to **create/delete/rename entries**

So 400 gives only r, but **without x you can't actually access the directory.**

2.

Observation table :

Permission	Regular File – Test	Observation	Directory – Test	Observation
200 (-w-)	Read and modify	Read denied; modification allowed	List and enter, create	List and enter denied; cannot create through directory

```

(tanu@kali)-[~]
$ ls sample.txt
sample.txt

(tanu@kali)-[~]
$ cat sample.txt
cat: sample.txt: Permission denied

(tanu@kali)-[~]
$ echo "hello" >> sample.txt

(tanu@kali)-[~]
$ cta sample.txt
Command 'cta' not found, did you mean:
  command 'cba' from deb cba
  command 'cha' from deb chawan
  command 'cat' from deb coreutils
  command 'ctr' from deb containerd
Try: sudo apt install <deb name>

(tanu@kali)-[~]
$ cat sample.txt
cat: sample.txt: Permission denied

(tanu@kali)-[~]
$ chmod 400 sample_dir

(tanu@kali)-[~]
$ cat sample.txt
cat: sample.txt: Permission denied

(tanu@kali)-[~]
$ chmod 400 sample.txt

(tanu@kali)-[~]
$ cat sample.txt
hello

```

For directory:

```

(tanu@kali)-[~]
$ chmod 200 sample_dir

(tanu@kali)-[~]
$ ls -ld sample_dir
d-w----- 2 tanu tanu 4096 Aug 21 00:10 sample_dir

(tanu@kali)-[~]
$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(tanu@kali)-[~]
$ cd sample_dir
cd: permission denied: sample_dir

(tanu@kali)-[~]
$

```

3.

Observation table :

Permission	Regular File – Test	Observation	Directory – Test	Observation
100 (--x).	Read and execute	Read denied; Execute permission is present, but the plain text file is not a valid executable program.	List and enter,	Listing denied; entering allowed

Screenshots:

```
(tanu@kali)-[~]
$ chmod 100 sample.txt

(tanu@kali)-[~]
$ ls -l sample.txt
--x----- 1 tanu tanu 6 Aug 21 00:35 sample.txt

(tanu@kali)-[~]
$ cat sample.txt
cat: sample.txt: Permission denied

(tanu@kali)-[~]
$ ./sample.txt
zsh: permission denied: ./sample.txt

(tanu@kali)-[~]
$
```

```
(tanu@kali)-[~]
$ chmod 100 sample_dir

(tanu@kali)-[~]
$ ls -ld sample_dir
d--x----- 2 tanu tanu 4096 Aug 21 00:10 sample_dir

(tanu@kali)-[~]
$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(tanu@kali)-[~]
$ cd sample_dir

(tanu@kali)-[~/sample_dir]
$ ls
ls: cannot open directory '.': Permission denied

(tanu@kali)-[~/sample_dir]
$
```

4.

Observation table :

Permission	Regular File – Test	Observation	Directory – Test	Observation
500 (r-x).	Read and modify	Read allowed; modification denied	List and enter,	Listing and entering allowed

Screenshots:

```
(tanu@kali)-[~]
$ chmod 500 sample.txt

(tanu@kali)-[~]
$ ls -l sample.txt
-r-x----- 1 tanu tanu 6 Aug 21 00:35 sample.txt

(tanu@kali)-[~]
$ cat sample.txt
hello

(tanu@kali)-[~]
$ echo "world" >> sample.txt
zsh: permission denied: sample.txt
```

For directory :

```
(tanu@kali)-[~]
$ chmod 500 sample_dir

(tanu@kali)-[~]
$ ls -ld sample_dir
dr-x----- 2 tanu tanu 4096 Aug 21 00:10 sample_dir

(tanu@kali)-[~]
$ ls sample_dir
file1.txt

(tanu@kali)-[~]
$ cd sample_dir

(tanu@kali)-[~/sample_dir]
$ touch newfile.txt
touch: cannot touch 'newfile.txt': Permission denied
```

5.

Observation table :

Permission	Regular File – Test	Observation	Directory – Test	Observation
600 (rw-)	Read and modify	Read and modification allowed	List, enter, create	Listing, entering and creating denied

Screenshots:

```
(tanu@kali)-[~]
$ chmod 600 sample.txt

(tanu@kali)-[~]
$ ls -l sample.txt
-rw----- 1 tanu tanu 6 Aug 21 00:35 sample.txt

(tanu@kali)-[~]
$ cat sample.txt
hello

(tanu@kali)-[~]
$ echo "world" >> sample.txt

(tanu@kali)-[~]
$ cat sample.txt
hello
world
```

For directory:


```
(tanu@kali)-[~]
$ chmod 600 sample.txt

(tanu@kali)-[~]
$ ls -l sample.txt
-rw----- 1 tanu tanu 6 Aug 21 00:35 sample.txt

(tanu@kali)-[~]
$ cat sample.txt
hello

(tanu@kali)-[~]
$ echo "world" >> sample.txt

(tanu@kali)-[~]
$ cat sample.txt
hello
world

(tanu@kali)-[~]
$ chmod 600 sample_dir

(tanu@kali)-[~]
$ ls -ld sample_dir
drw----- 2 tanu tanu 4096 Aug 21 00:10 sample_dir

(tanu@kali)-[~]
$ ls sample_dir
ls: cannot access 'sample_dir/file1.txt': Permission denied
file1.txt

(tanu@kali)-[~]
$ cd sample_dir
cd: permission denied: sample_dir

(tanu@kali)-[~]
$ touch sample_dir/newfile.txt
touch: cannot touch 'sample_dir/newfile.txt': Permission denied
```

6.

Observation table :

Permission	Regular File – Test	Observation	Directory – Test	Observation
700 (rwx)	Read, modify and execute	Read and modification allowed; execute permission present, but plain text file is not executable	List, enter, create and delete	All operations allowed

Screenshots:

```
(tanu@kali)-[~]  
$ chmod 700 sample.txt  
  
(tanu@kali)-[~]  
$ ls -l sample.txt  
-rwx----- 1 tanu tanu 12 Aug 21 01:01 sample.txt  
  
(tanu@kali)-[~]  
$ cat sample.txt  
hello  
world  
  
(tanu@kali)-[~]  
$ echo "modified" >> sample.txt  
  
(tanu@kali)-[~]  
$ ./sample.txt  
./sample.txt: 1: hello: not found  
./sample.txt: 2: world: not found  
./sample.txt: 3: modified: not found
```

For directory:

```
(tanu@kali)-[~]
$ chmod 700 sample.txt

(tanu@kali)-[~]
$ ls -l sample.txt
-rwx----- 1 tanu tanu 12 Aug 21 01:01 sample.txt

(tanu@kali)-[~]
$ cat sample.txt
hello
world

(tanu@kali)-[~]
$ echo "modified" >> sample.txt

(tanu@kali)-[~]
$ ./sample.txt
./sample.txt: 1: hello: not found
./sample.txt: 2: world: not found
./sample.txt: 3: modified: not found

(tanu@kali)-[~]
$ chmod 700 sample_dir

(tanu@kali)-[~]
$ ls -ld sample_dir
drwx----- 2 tanu tanu 4096 Aug 21 00:10 sample_dir

(tanu@kali)-[~]
$ ls sample_dir
file1.txt

(tanu@kali)-[~]
$ cd sample_dir

(tanu@kali)-[~/sample_dir]
$ touch newfile.txt
```

```
(tanu@kali)-[~/sample_dir]
$ touch newfile.txt

(tanu@kali)-[~/sample_dir]
$ ls
file1.txt newfile.txt

(tanu@kali)-[~/sample_dir]
$ rm newfile.txt

(tanu@kali)-[~/sample_dir]
$ ls
file1.txt

(tanu@kali)-[~/sample_dir]
$
```

Questions:

1. What does read (r) permission allow you to do with a regular file?

Read permission allows the user to view or read the contents of a regular file.

For example:

```
cat sample.txt
```

2. What does read (r) permission allow you to do with a directory?

Read permission allows the user to read the list of entries, or filenames, contained in the directory.

For example:

```
ls sample_dir
```

However, directory read permission normally needs x permission as well for useful access to the entries.

3. What does write (w) permission mean for a regular file?

Write permission allows the user to modify the contents of the file.

For example:

```
echo "hello" >> sample.txt
```

4. What does write (w) permission mean for a directory?

Write permission on a directory allows modification of its directory entries, such as creating, deleting, or renaming files.

However, x permission is also required for normal access to the directory entries.

5. What does execute (x) permission mean for a regular file?

Execute permission allows a file to be executed as a program or script, provided the file contains valid executable code.

6. Why is execute (x) permission important for accessing a directory?

Execute permission on a directory allows the user to enter and search/access the directory.

Without x permission, the user cannot normally access the directory's entries even if read permission is present.

7. Which permission combination allows the owner to fully access a regular file?

700 (rwx) gives the owner read, write, and execute permissions.

8. Which permission combination allows the owner to fully access a directory?

700 (rwx) gives the owner read, write, and execute permissions, allowing the owner to list, access, create, delete, and modify entries in the directory.

9. Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.

The meaning of permissions depends on the type of object.

For a **regular file**:

- r means read the file contents.
- w means modify the file contents.
- x means execute the file.

For a **directory**:

- r means read the list of directory entries.
- w means create, delete, or rename entries.
- x means enter/search/access the directory.

Therefore, the same permission letters have different effects because a file contains data, while a directory contains and manages references to other files and directories.

Although the same r, w, and x symbols are used for both, their meanings are different depending on the object. For files, the permissions control reading, modifying, and executing the file. For directories, they control listing entries, modifying entries, and accessing/searching the directory.